

C 1 38°43' 2 27°42' 3 84°39' 4 132°30' 5 52°32'
6 20°27' 7 44°25' 8 77°20' 9 96°15' 10 31°21'
11 146°33' 12 88°36' 13 102°50' 14 112°53'
15 73°42'

9 The cosine rule: mixed questions

A 1 11.5 cm 2 10.4 cm 3 14.0 cm 4 43.2 cm
5 22.8 m 6 6.6 m 7 62° 8 86° 9 41° 10 39°
11 53° 12 64°
B 1 17.6 cm 2 11.7 cm 3 3.7 m 4 11.4 m 5 77.2 mm
6 9.1 m 7 111°38' 8 23°47' 9 84°55' 10 32°31'
11 87°15' 12 48°51'
C 1 9.1 cm 2 19.5 cm 3 9.9 m 4 39.7 m 5 55°4'
6 126°25' 7 109°29' 8 69°31'

10 Mixed questions: the sine and cosine rules

A 1 7.4 cm 2 29.1 cm 3 28.1 cm 4 10.5 cm
5 43.0 cm 6 185.0 cm 7 15.1 m 8 54.1 cm
9 201.5 mm
B 1 39° 2 57° 3 31° 4 40° 5 48° 6 63° 7 45° 8 25°
9 12°
C 1 8.1 m 2 57°30' 3 18°43' 4 20.4 cm 5 67°57'
6 72°58' 7 82°54' 8 29.5 m 9 8.5 m 10 60°0'
11 105°49' 12 7.5 m

11 Problems involving two triangles

A 1 10.5 cm 2 31.6 cm 3 25.8 cm 4 71° 5 54° 6 33°
B 1 18.3 m 2 BD = 129.1 m, CD = 107.1 m 3 81 m
4 AB = 60.7 m, BD = 36.5 m 5 44.7 m 6 155 m
7 BD = 37.1 m, BC = 48.9 cm 8 BD = 18.6 cm,
A = 91°
C 1 267.1 m, 185.5 m 2 3556 m 3 132 cm, 146 cm
4 274.2 m 5 35.5 cm 6 63°

12 Review: the sine rule and the cosine rule

A 1 20.8 cm 2 23.9 cm 3 27.1 cm 4 45° 5 60°
6 125° 7 18.6 cm 8 22.0 cm 9 13.1 cm 10 47°
11 48° 12 97°
B 1 3.6 m 2 5.6 m 3 2.6 m 4 52°43' 5 112°2'
6 50°11' 7 141 mm 8 124 mm
C 1 9.1 cm 2 22.6 cm 3 27.0 cm 4 13.0 cm
5 35°2' 6 63°36' 7 58°46' 8 119°30' 9 47°
10 32 km 11 5.2 m 12 51°12'

13 The area of a triangle

A 1 19° 2 75° 3 60° 4 75° 5 88° 6 40° 7 53° 8 42°
9 1.563 10 7.502 11 9.798 12 71.629
B 1 194.9 cm² 2 145.9 cm² 3 37.9 cm² 4 113.8 cm²
5 1107.6 cm² 6 474.1 cm² 7 13.4 cm² 8 100.4 cm²
9 42.0 m² 10 7.2 m² 11 138.5 m² 12 222.2 m²
C 1 68.52 cm² 2 92.25 cm² 3 40.14 m² 4 5.53 m²
5 0.87 m² 6 8.05 m² 7 4115.71 m² 8 95.41 m²
9 27.80 m² 10 114.37 m² 11 63.26 m² 12 14.61 m²
13 291.84 m² 14 10.57 m² 15 6.26 m²

14 Practical application: radial surveys

A 1 1.7 m 2 1.7 m 3 2.7 m 4 73° 5 47° 6 53°
7 4.3 m² 8 8.0 m² 9 3.2 m²
B 1 a 85.4 m b BC = 91.9 m, CD = 36.9 m, DA = 74.2 m
c 288 m d 1495.9 m² e BPC = 1041.1 m², CPD =
755.0 m², APD = 1322.2 m² f 4615 m² 2 perimeter =
385 m, area = 8975 m² 3 perimeter = 348 m, area =
7142 m²
C 1 a 115° b 87° c 101.3 m d 331 m e 6669 m²
2 perimeter = 385 m, area = 9003 m² 3 perimeter =
347 m, area = 7496 m²

15 Mixed problems

A 1 19.8 cm 2 41.4 cm 3 25.4 cm 4 43° 5 116°
6 108° 7 323.1 cm² 8 332.3 cm² 9 562.7 cm²
B 1 42 m² 2 15° 3 20.5 km 4 2.1 m 5 53 cm 6 122°
7 3402 cm² 8 19.1 km
C 1 91 mm 2 166 m 3 791.5 m 4 23.4 cm² 5 28.6 km
6 60°21' 7 8.2 km 8 172 873 mm² 9 26.4 n. miles
10 207°

16 The graph of $y=\sin x^\circ$

A 1 0, 0.5, 0.87, 1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87,
-0.5, 0
0, -0.5, -0.87, -1, -0.87-0.5, 0, 0.5, 0.87, 1, 0.87, 0.5, 0
B 1 1 2 0 3 -1 4 270° 5 30°, 150° 6 90° 7 60°,
120° 8 0, 180°, 360° 9 210°, 330°
C 4 a $y = -2\sin x$ b $y = \sin 2x$ c $y = 3\sin x$
d $y = \sin \frac{1}{2}x$ 5 3 6 360° 7 360° 8 120° 9 -1

17 The graph of $y=\cos x^\circ$

A 1 1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87, -0.5, 0, 0.5,
0.87, 1
1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87, -0.5, 0, 0.5, 0.87, 1

B 1 1 2 0 3 0 4 0°, 360° 5 60°, 300° 6 180° 7 0.5
8 -0.5 9 0.5
C 1 1, 0.87, 0.71, 0.5, 0, -0.5, -0.71, -1, -0.71, 0, 1
2 2, 1.73, 1.41, 1, 0, -1, -1.41, -2, -1.41, 0, 2
3 1, 0.5, 0, -0.5, -1, -0.5, 0, 1, 0, -1, 1
4 -1, -0.87, -0.71, -0.5, 0, 0.5, 0.71, 1, 0.71, 0, -1
5 a $y = 2\cos x$ b $y = \cos 2x$ c $y = \cos \frac{1}{2}x$
d $y = -\cos x$ 6 1 7 360° 8 3 9 180°

18 The graph of $y=\tan x^\circ$

A 1 0, 0.58, 1, 1.73, undefined, -1.73, -1, -0.58, 0, 0.58,
1, 1.73, undefined, -1.73, -1, -0.58, 0
B 1 continuous 2 -270°, -90°, 90°, 270° 3 180°, 180°
4 amplitude 5 asymptotes 6 -360°, -180°, 0°, 180°,
360° 7 -315°, -135°, 45°, 225° 8 -225°, -45°, 135°,
315°
C 1 180° 2 $x = 90^\circ$, $x = 270^\circ$ 3 60° 4 $x = -30^\circ$,
 $x = 30^\circ$, $x = 90^\circ$ 5 0, 3, -3, 0, 3, -3, 0 6 0, 1,
undefined, -1, 0, 1, undefined, -1, 0 7 $y = \tan 2x$

19 Further trigonometric graphs

A 1 $a = 0.866$, $b = 90^\circ$, $c = 0.5$, $d = -0.5$, $e = 330^\circ$, $f = 0$
3 a $x = 45^\circ$, 135° b $x = 225^\circ$, 315° c $x = 210^\circ$, 330°
d $x = 0^\circ$, 180° , 360°
B 1 a 90°, 270° b infinite c 107°, 287° d 215°
2 A(90°,3), B(180°,0), C(270°,-3), D(360°,0) 3 $p = 3$,
 $q = 1$, $m = 2$, $n = \frac{1}{2}$ ($y = 3\sin x$ and $y = 2\cos \frac{1}{2}x$);
 $y = 3\sin x$ has period 360°, amplitude = 3;
 $y = 2\cos \frac{1}{2}x$ has period 720°, amplitude = 2
C 2 $\cos x$: 1, 0.5, 0, -0.5, -1, -0.5, 0, 0.5, 1
 $\sec x$: 1, 2, undefined, -2, -1, -2, undefined a 360°
b infinite c infinite
3 $\sin x$: 0, 0.5, 1, 0.5, 0, -0.5, -1, -0.5, 0
 $\operatorname{cosec} x$: undefined, 2, 1, 2, undefined, -2, -1, -2,
undefined a 360° b infinite c infinite

20 Simple trigonometric equations

A 1 37° 2 67° 3 22° 4 25° 5 73° 6 41° 7 57°
8 30° 9 39° 10 58° 11 67° 12 44° 13 48° 14 51°
15 57° 16 89° 17 73° 18 84° 19 69° 20 6° 21 67°
22 62° 23 32° 24 74° 25 28° 26 25° 27 19° 28 29°
B 1 143° 2 166° 3 121° 4 175° 5 102° 6 115°
7 139° 8 99° 9 153° 10 129° 11 117° 12 108°
13 30°, 150° 14 44°, 136° 15 59°, 121° 16 13°, 167°
17 65° 18 120° 19 136° 20 30° 21 27° 22 124°
23 68° 24 126° 25 24°, 156° 26 60° 27 45° 28 53°,
127° 29 107° 30 117° 31 30°, 150° 32 72° 33 48°

34 112° 35 120° 36 49°, 131°
C 1 60°, 300° 2 210°, 330° 3 56°, 236° 4 135°, 315°
5 117°, 297° 6 229°, 311° 7 58°, 238° 8 201°, 338°
9 38°, 322° 10 120°, 300° 11 108°, 208° 12 233°,
307° 13 90° 14 63°, 243° 15 180° 16 41°, 319°
17 270° 18 50°, 230° 19 42°, 138° 20 27°, 207°
21 71°, 289° 22 127°, 307° 23 30°, 150° 24 124°,
304°

21 Trigonometric relationships

A 1 true 2 true 3 true 4 true 5 true 6 true 7 true
8 true

22 Review: trigonometry

A 1 9.9 cm 2 6.6 cm 3 33.9 cm 4 44°54' 5 34°23'
6 54°19' 7 31.9 cm 8 31.3 cm 9 31.1 cm 10 82°
11 61° 12 40° 13 136 cm² 14 321 cm² 15 234 cm²
B 1 100° 2 103° 3 118° 4 3.8 m 5 4.7 m 6 1.9 m
7 91°28' 8 58°16' 9 93°12' 10 902 cm² 11 401 m
12 217 mm 13 347.4 m 14 6924 mm
15 AB = 56.0 m, BD = 37.5 m
C 1 23.9 cm 2 27.6 cm 3 19.3 cm 4 40.5 cm 5 45°11'
6 54°31' 7 35°5' 8 116°48' 9 245 mm 10 77°3'
11 186.7 m 12 perimeter = 298 m, area = 5404 m²
13 perimeter = 348 m, area = 7142 m²